Course Syllabus
CS -390 Introduction to Theoretical Computer Science

Course Information

- **Instructor(s):** Dr. Sumaya Sanober
- **Course title, number, section:** CS 390 – Introduction to Theoretical Computer Science
- **Semester and year: Fall Semester, 2026**
- **Delivery mode (synchronous or asynchronous online):** asynchronous online

OLD DOMINION UNIVERSITY

Course Description

Catalog Description

Elementary study of theoretical aspects of computer science. Topics in formal languages and automata theory are covered including regular languages, regular expressions, finite automata, context-free languages, pushdown automata, grammar, Turing machines, and unsolvable problems.

Course Description

This is an elementary study of theoretical aspects of computer science. Topics in formal languages and automata theory are covered including regular languages, regular expressions, finite automata, context-free languages, pushdown automata, grammar, Turing machines, and unsolvable problems. This is a web-based class requiring independent responsibility and online communication skills on the part of the student. There are no regularly scheduled class meetings.

Prerequisites

A grade of C or better in CS 381 and in any one of: CS 250, CS 251, CS 253, ECE 250. Students are also expected to be familiar with the use of standard Internet-based tools including email and web browsers.

Required Materials

Required Textbook

Hopcroft, John E. and Motwani, Rajeev and Ullman, Jeffrey D., Introduction to Automata Theory, Languages, and Computation (3rd Edition), 2006, ISBN 0321462254, Addison-Wesley Longman Publishing Co.

Technology Requirements

- Hardware: [i.e., microphone, speakers, webcam, etc.]
- Software: [i.e., Microsoft 365, Adobe Acrobat, etc.]

Course Learning Objectives

- CLO1: Demonstrate knowledge of the primary mathematical models of computation (automata) that underpin Computer Science.
- CLO2: Demonstrate knowledge of formal languages and their relationship to automata

- CLO3: Be able to construct, work with, and prove properties of the basic forms of automata and formal languages.
- CLO4: Understand and be able to discuss the relationship of automata and formal languages to practical issues in programming languages and software development.

Grading Criteria

In this course, I will calculate your final grade based on the following categories:

Assignments	Percentage/Points
Quizzes	50%
Checkpoint	5%
Midterm Exam	15%
Final Exam	30%

All letter grades are based on ODU's 4-point scale.

Grading Scale

Letter grades are based on the following percentage scale:

Percentage	Letter Grade
A	4.0
A-	3.7
B+	3.3
B	3.0
B-	2.7
C+	2.3
C	2.0
C-	1.7
D+	1.3
D	1.0
D-	0.7
F	0.7

All grades for assignments and exams will be recorded as a number on this 4-point scale. Intermediate values (e.g., 3.5) falling between letter grades are possible.

Although 4.0 is the largest value on the official ODU scale, individual grades on assignments and exams may range as high as 4.5.

Converting Raw Grades to 4-pt Scale

Scores on assignments and exams will first be converted to a percentage based on the number of points possible on that item.

Percentages are then mapped to the 4-point scale as follows:

Percentage	4-Point Scale
100	4.5
≥ 92	4.0
≥ 76	3.0
≥ 60	2.0
≥ 44	1.0
≤ 27	0.0

Policy on Incomplete Grades

A grade of "I" indicates assigned work yet to be completed in a given course, or absence from the final examination, and is assigned only upon instructor approval of a student request. The "I" grade may be awarded only in exceptional circumstances beyond the student's control. The "I" grade becomes an "F" if not removed by the day grades are due for the following term, based on specific criteria: Incomplete, Withdraws and Z grades.

Assignment Requirements

Assignment Category: Weekly Quizzes – 50%

Purpose:

The weekly quizzes are designed to reinforce foundational and progressive understanding of automata theory and formal languages. They ensure continuous engagement and help students build fluency in constructing and analyzing computational models in an accelerated format.

Course Learning Objectives:

- CLO1: Demonstrate knowledge of the primary mathematical models of computation (automata)
- CLO2: Demonstrate knowledge of formal languages and their relationship to automata
- CLO3: Construct, analyze, and prove properties of automata and formal languages

Frequency:

- Occurs in Modules 1–8 (weekly)
- Due Sunday at 11:59 PM

Instructions:

1. Review assigned readings, lecture materials, and worked examples.
2. Complete the quiz in Canvas within the allotted time.
3. Questions may include:
    - Conceptual definitions (automata, languages)
    - DFA/NFA construction
    - Language recognition and analysis
4. Ensure all answers are your own work.

Deliverables:

- Canvas quiz submission (auto-graded and/or instructor-reviewed)

Evaluation Criteria:

- Accuracy of responses
- Correct application of theoretical concepts
- Logical consistency in problem-solving

Assignment Category: Checkpoint Assessments – 5%

(Checkpoint 1 – Module 3 | Checkpoint 2 – Module 5)

Purpose:

Checkpoint assessments evaluate cumulative understanding at key transition points (Automata → Regular Languages). These structured assessments ensure students can integrate concepts before advancing in the accelerated course.

Course Learning Objectives:

- CLO2: Demonstrate knowledge of formal languages and their relationship to automata
- CLO3: Construct, analyze, and prove properties of automata and formal languages

Frequency:

- 2 total checkpoints

- Module 3: Automata (DFA/NFA + equivalence)
- Module 5: Regular Languages (closure, equivalence)

Instructions:

1. Solve multi-step theoretical problems.
2. Clearly show all work, including:
    - Automata construction
    - Conversions (NFA → DFA, RegEx ↔ Automata)
    - Proof-based reasoning where required
3. Use proper notation and formatting.

Deliverables:

- Typed or neatly scanned submission (PDF/Word) via Canvas

Evaluation Criteria:

- Completeness of solutions
- Correctness of constructions and transformations
- Clarity and rigor of reasoning

Assignment Category: Midterm Exam – 15%

(Covers Modules 1–4: Foundations → Regular Expressions)

Purpose:

The midterm exam evaluates students understanding of core automata concepts and their relationship to formal languages before progressing to more advanced computational models.

Course Learning Objectives:

- CLO1: Demonstrate knowledge of automata models
- CLO2: Demonstrate knowledge of formal languages
- CLO3: Construct and analyze automata

Frequency:

- Conducted in Module 4

Instructions:

1. Complete the proctored exam using the approved proctoring system.
2. Exam components may include:
    - DFA/NFA design and analysis
    - NFA → DFA conversion
    - Regular expression construction

3. Provide clear and complete solutions.

Deliverables:

- Canvas-based exam submission

Evaluation Criteria:

- Accuracy of solutions
- Correct use of formal notation
- Depth of conceptual understanding

Assignment Category: Final Exam – 30%

Purpose:

The final exam assesses mastery of advanced models of computation and their theoretical limits, including grammars, pushdown automata, and Turing machines.

Course Learning Objectives:

- CLO1: Demonstrate knowledge of advanced computation models
- CLO3: Construct and proof of automata and languages
- CLO4: Understand relationships with real-world computing systems

Frequency:

- Conducted in Module 8

Instructions:

1. Complete the proctored comprehensive exam.
2. Topics include:
    - Context-Free Grammars (CFGs)
    - Pushdown Automata (PDA)
    - Turing Machines and decidability
3. Show all reasoning and intermediate steps clearly.

Deliverables:

- Canvas exam submission

Evaluation Criteria:

- Conceptual depth and correctness
- Proper construction of models
- Ability to synthesize across topics

Assignment Category: Participation & Self-Assessments – 0% (Optional/Low-Stakes Support)

Purpose:

Supports student learning through self-reflection, practice, and engagement in an asynchronous environment.

Course Learning Objectives:

- CLO2: Strengthen understanding of formal languages
- CLO3: Reinforce problem-solving and reasoning skills

Frequency:

- Embedded across all modules

Instructions:

1. Complete self-assessment and practice exercises.
2. Engage in optional discussions or problem walkthroughs.

Deliverables:

- Canvas activity completion

Evaluation Criteria:

- Completion-based
- Engagement and effort

Submission & Policy Notes

- All assignments must be submitted via Canvas LMS
- Late Work Policy:
    - Quizzes: No late submissions (lowest score dropped)
    - Checkpoints: 10% penalty per day (up to 48 hours)
    - Exams: No late submissions without documented emergency
- Students must adhere to academic integrity policies; all work must be original

Expectations (Optional)

What you can expect from me, as your instructor:

- I will ensure the accuracy of the course content.
- I will provide clear guidelines regarding the course assignments.
- I will be available to answer your questions.
- I will provide meaningful feedback in a timely manner.
- I will do my best to create a worthwhile learning experience for all my students.
- I will follow fair and clear evaluation guidelines for all the course assignments.

- I will do my best to prepare you for exams and assignments.

What I expect from you, as my student:

- You will complete all course readings and assignments in a timely manner.
- You will follow the ODU honor pledge.
- You will follow proper netiquette.
- You will interact with your faculty and classmates professionally and respectfully.
- You will avoid sarcasm and inappropriate language, including the use of ALL CAPS.
- You will acknowledge your classmates' ideas and build on them to contribute to the discussion.

Office Hours:

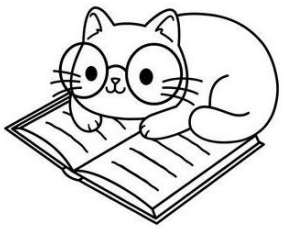
For Zoom Office hours: book here, Sumaya Sanober, ODU CS Dept. Twice a week, on Tuesday (11:00 AM-12:00 PM) and Thursday (12:00 PM-1:00 PM)

- General questions about course content and reports of website problems should normally be asked in the Discussions area in Canvas or via email.
- Questions about grades, quiz content, and other graded activities **must** be sent to ssanober@odu.edu , not posted in the Discussion area.

Communication

- Communication Expectations: Clear communication is one of the most important skills in computer science not only in this course, but also in industry, research, and professional teamwork. In CS390, asking thoughtful questions, explaining ideas clearly, and communicating respectfully will help you succeed both academically and professionally. The course includes several ways to communicate, including email, discussion boards, announcements, and office hours.
- Professional Communication Matters: As future computing professionals, students are expected to communicate with professionalism, clarity, and respect in all course interactions. Whether you are sending an email, participating in discussions, or asking for help, strong communication habits make collaboration more effective and help you receive faster and more meaningful support.
- Asking Good Questions: One of the most valuable skills in computer science is learning how to ask precise and well-prepared questions. When requesting help, provide enough detail about:

- what you attempted,
- where you are stuck,
- what errors or results you encountered,
- and the exact messages or outputs you received.
- The more specific your question is, the easier it becomes to diagnose the issue and guide you toward a solution. In computing, details matter vague questions often lead to delayed or incomplete answers.
- Discussion Boards and Email Etiquette: Discussion boards are intended for course-related clarification questions and collaborative learning, while private matters such as grades or assignment-specific solutions should remain in private communication with the instructor. Students are expected to follow professional online etiquette and maintain a respectful learning environment for everyone.
- A Quick Reminder: In computer science, communication is not separate from technical skill it is part of the skill. Great programmers, researchers, and engineers are also great communicators. Learning how to explain problems clearly, collaborate respectfully, and engage thoughtfully will benefit you far beyond this course.



Critical Reading**:** Critical Reading: CS 390 is a theory-intensive course that involves a significant amount of reading and careful analysis of concepts, examples, proofs, and problem-solving techniques. Success in this course requires students to actively engage with the reading materials on a regular basis rather than relying only on lectures or videos. Many topics in automata theory and formal languages build upon one another, so critical reading and consistent review are essential for understanding the material and keeping pace with the course. Students are expected to read course notes, examples, and support materials carefully before and after class activities.

Course Policies

Attendance and Participation

Required Software

CS Account

All students taking this course must have activated a login and e-mail account on the CS Dept.'s Unix network. (This is distinct from any Midas or other account you may have from the general University computer center the ODU ITS). You may have a CS account already if you were registered for a CS class in a recent semester. If not, you will need to create a new account. Go to the CS Dept. home page and look for "Account Creation". All students on this course are responsible for making sure they have a working CS Unix account by the end of the first week of the semester.

Web Browser

You will need a reasonably up-to-date version of the Edge, Firefox, or Chrome web browser. Other browsers or older versions of these may also be acceptable, but cannot be guaranteed so, because the course materials are not tested with other and older browsers. Neither Internet Explorer (Windows) nor Safari (Apple) are recommended. They might work. They might not. You will need to permit JavaScript to run in your browser.

Exam Proctoring

Exam Proctoring

The midterm and final exams will be proctored. You will have a variety of proctoring options available to you via Smart Proctoring.

•All students will be responsible for arranging their proctoring choices at least one week in advance of each exam.

The weekly quizzes are not proctored.

Late Assignments

Late quizzes and make-up exams will not normally be permitted.

Exceptions to this and other grading policies will be made only in situations of unusual and unforeseeable circumstances beyond the student's control, and such arrangements must be made promptly, prior to the due date in any situation where conflict is foreseeable.

"I've fallen behind and can't catch up", "I'm having a busier semester than I expected", or "I registered for too many classes this semester" are not grounds for an extension

Academic Honesty

Everything turned in for grading in this course must be your own work. Quizzes are open books, open-notes open-course website. During exams you may access the course website. You may not, however, use non-course websites nor consult with another human, either in person, or electronically, on exams and quiz material. You are expected to conform to academic standards in external resources, below. avoiding plagiarism. See also the discussion on using the instructor reserves the right to question a student orally or in writing and to use his evaluation of the student's understanding of the assignment and of the submitted solution as evidence of cheating. Students who contribute to violations by sharing their solutions with others may be found to be in violation of this policy. This includes showing material to other students in person and posting partial, complete, or even speculative solutions in any public area, whether physical or on the internet.

This policy is not intended to prevent students from providing legitimate assistance to one another. Students are encouraged to seek/provide one another aid in learning to use the support systems, or to general issues relating to the course subject matter. The same guideline applies to discussions, whether face-to-face or on-line, with anyone other than the course instructor and TAs – general aid on the subject matter of the course is OK. Specific discussions of solutions to any graded activity are forbidden. Violations will be reported to the Office of Student Conduct and Academic Integrity.

Using External Resources

In quizzes and exams, you may use "answers" that you find already posted on the internet (but may not solicit new ones) if you acknowledge your sources appropriately.

If you use someone else's thoughts, proofs, or arguments, you must cite your source appropriately, in a fashion that allows me to verify it.

If you use someone else's wording, you must enclose that wording in quotation marks and cite your source.

That said, you are responsible for making sure that the cited and/or quoted material is relevant to the question I asked, even when taken out of context from the original source.

Nothing shows me that you do not understand the course material faster than copying an answer to a different question than the one I asked!

Many of the lowest scores on prior semesters' exams came from students copying and pasting answers that only matched the question I asked in a few key words or phrases.

I have low patience for this. If a clear copied answer is not obviously relevant to the question I asked, I am not going to spend a lot of effort trying to study it and figure out ways in which it might be twisted into relevance.

Therefore,

You are responsible for any alterations or bridging discussions necessary to make a copied argument or quotation match the question being asked.

If the quoted source uses different notation schemes than what we have introduced in this course, you must reformulate the notation into that of this course.

Be aware that different authors may have subtle differences in definitions of terminology, statement of theorems, etc. Again, if you quote external material, it is your responsibility to recognize and adjust for such differences.

Course Disclaimer

The course schedule and activities are subject to change. Changes will be posted as Announcements in Canvas.

AI Usage (Optional)

This course encourages students to develop independent problem-solving and proof-writing skills. You may consult external resources, including textbooks, websites, discussion forums, and AI-based tools (such as ChatGPT or other generative AI systems), as study aids unless a specific assignment or assessment explicitly prohibits them.

However, the following expectations apply:

If you use ideas, explanations, proofs, examples, or arguments from another source, you must clearly acknowledge and cite that source appropriately.

If you directly copy wording from a source, the copied material must appear in quotation marks with proper citation.
You are responsible for ensuring that any material you use is:
Relevant to the actual question being asked
Mathematically and logically correct
Consistent with the notation, terminology, and definitions used in this course

AI-generated content is not always correct. AI systems may produce incorrect proofs, invalid reasoning, misleading explanations, or notation inconsistent with course material. You are responsible for verifying the accuracy of any AI-assisted work you submit.

Simply copying and pasting solutions from the internet or AI tools without understanding them will likely result in low scores, especially in proof-based questions where reasoning, clarity, and correctness are essential.

If an external source uses notation or terminology different from what we use in class, you must rewrite the material using the notation and conventions introduced in this course.

During quizzes and exams, follow the specific instructions provided for that assessment. Some assessments may restrict or prohibit the use of AI tools or external resources.

Remember: the goal of this course is not only obtaining answers, but learning how to reason formally, construct valid arguments, and communicate computational ideas clearly.

University Policies

Code of Student Conduct and Academic Integrity

The Office of Student Conduct & Academic Integrity (OSCAI) oversees the administration of the student conduct system, as outlined in the Code of Student Conduct. Old Dominion University is committed to fostering an environment that is: safe and secure, inclusive, and conducive to academic integrity, student engagement, and student success. The University expects students and student organizations/groups to uphold and abide by standards included in the Code of Student Conduct. These standards are embodied within a set of core values that include personal and academic integrity, fairness, respect, community, and responsibility.

Honor Pledge

By attending Old Dominion University, you have accepted the responsibility to abide by the Honor Pledge:

I pledge to support the honor system of Old Dominion University. I will refrain from any form of academic dishonesty or deception, such as cheating or plagiarism. I am aware that as a member of the academic community it is my responsibility to turn in all suspected violations of the Honor Code. I will report to a hearing if summoned.

Discrimination Policy

The purpose of this policy is to establish uniform guidelines in order to promote a work and education environment that is free from harassment and discrimination, as defined below, and to affirm the University's commitment to foster an environment that emphasizes the dignity and worth of every member of the Old Dominion University community. The Discrimination Policy details the process to address complaints or reports of retaliation, as defined by this policy.

Diversity and Inclusion

The Division of Student Engagement & Enrollment Services values the uniqueness of our Monarch community. The word "engagement" reflects our commitment to embrace the differences in our cultural backgrounds, perceptions, beliefs, traditions, world views, socio-economic status, cognitive and physical abilities.

We will strive to serve as the pre-eminent model for engaging every student to achieve their own success. Our core values are fueled by our responsibility and actions toward community development and engagement, cultural competence and understanding, physical and mental wellness and inclusion for every member of ODU. We will embrace our greatest strength - the diverse composition of our student body and workforce. For more information regarding diversity and inclusion, please visit the Office of Intercultural Relations.

Educational Accessibility and Accommodations

Old Dominion University is committed to ensuring equal access to all qualified students with disabilities in accordance with the Americans with Disabilities Act. The Office of Educational Accessibility (OEA) is the campus office that works with students who have disabilities to provide and/or arrange reasonable accommodations.

The Accommodations for Students with Disabilities define the procedures used to accommodate student with disabilities. Students are encouraged to self-disclose disabilities that the Office of Educational Accessibility has verified by providing Accommodation Letters to their instructors early in the semester to start receiving accommodations. Accommodations will not be made until the Accommodation Letters are provided to instructors each semester.

University Email Policy

With the increasing reliance and acceptance of electronic communication, email is considered an official means for University communication. Old Dominion University provides each student an email account for the purposes of teaching and learning, research, administration, and service. It is the responsibility of every eligible student to activate MIDAS, the Monarch Identification and Authorization System, to obtain email access. It is important that all students are aware of the expectations associated with email use as outlined in the Student Email Standard. The email account provided by the University is an official point of contact for correspondence. Students are expected to check their official e-mail account on a frequent and consistent basis to stay current with university communications. Mail sent to the ODU email address may include notification of university-related actions, including academic, financial, and disciplinary actions. For more information about student email, please visit Student Computing.

Withdrawal

A syllabus constitutes an agreement between the student and the course instructor about course requirements. Participation in this course indicates your acceptance of its teaching focus, requirements, and policies. Please review the syllabus and the course requirements as soon as possible. If you believe that the nature of this course does not meet your interests, needs or expectations, if you are not prepared for the amount of work involved – or if you anticipate assignment deadlines or abiding by the course policies will constitute an unacceptable hardship for you – you should drop the course by the drop/add deadline, which is listed in the ODU Schedule of Classes. For more information, please visit the Office of the University Registrar.

Privacy of Student Information

Old Dominion University recognizes its duty to uphold the public's trust and confidence, not only in following laws and regulations, but in following high standards of ethical behavior. Members of the Old Dominion University community are responsible for maintaining the highest ethical standards and principles of integrity. The Code of Ethics is a set of values-based statements that demonstrate the University's commitment to this goal. The Privacy of Student Information details Family Educational Rights & Privacy Act (FERPA), along with other information regarding privacy.

Course Schedule

Week/Module	Date	Topics	Assignments	Due Date
Module 1	Aug 24 – Aug 30	Foundations & Mathematical Readiness (Sets, relations, functions, notation, course tools)	Quiz 1: Foundations & Math Review	Aug 30, 11:59 PM
Module 2	Aug 31 – Sept 6	Deterministic Finite Automata (DFA) (Formal definition, design, basic problems)	Quiz 2: DFA Design & Analysis	Sept 6, 11:59 PM
Module 3	Sept 7 – Sept 13	NFA + DFA Equivalence (NFA concepts, subset construction, equivalence proofs)	Quiz 3: NFA & Conversion Checkpoint 1: Automata Review	Sept 13, 11:59 PM
Module 4	Sept 14 – Sept 20	Regular Expressions & Conversions (Regex, automata conversion, pattern representation)	Quiz 4: Regular Expressions Midterm Exam (Modules 1–4)	Quiz: Sept 20, 11:59 PM Midterm: Sept 18–20
Module 5	Sept 21 – Sept 27	Regular Languages & Equivalence (Closure properties, DFA ⇄ NFA ⇄ RegEx equivalence)	Quiz 5: Regular Languages Checkpoint 2: Cumulative Regular Languages	Sept 27, 11:59 PM
Module 6	Sept 28 – Oct 4	Context-Free Grammars (CFGs) (Grammar rules, derivations, parse trees)	Quiz 6: CFGs	Oct 4, 11:59 PM
Module 7	Oct 5 – Oct 11	Pushdown Automata (PDA) + CFL Properties (PDA design, CFG ↔ PDA, pumping lemma, closure)	Quiz 7: PDA & CFL Properties	Oct 11, 11:59 PM

Module 8	Oct 12 – Oct 16	Turing Machines, Decidability & Church-Turing (TM models, undecidability, real-world connections)	Quiz 8: Turing Machines Final Exam (Comprehensive)	Quiz: Oct 16, 11:59 PM Final Exam: Oct 14–16

Module 1: Foundation & Mathematical Readiness

Module Overview
This module introduces course expectations, tools, and the mathematical foundations required for theoretical computer science. Students review sets, relations, functions, and the concept of formal languages, which underpin all later models of computation.

Module Learning Objectives (MLOs)
After completing this module, you will be able to:

- **MLO1:** Explain key discrete mathematics concepts including sets, relations, and functions (**CLO1**)
- **MLO2:** Interpret and construct formal definitions using mathematical notation (**CLO2**)
- **MLO3:** Describe the concept of a formal language and its role in computation (**CLO2**)

Required Readings & Resources

- Hopcroft et al., Chapter 1
- Lecture notes: Sets, Functions, Proof Techniques
- Self-assessment: Mathematics Review

Topics

- Sets, relations, functions
- Mathematical notation and proofs
- Strings and formal languages

Assignments

- Quiz 1: Foundations & Math | Aug 30
- Self-Assessment: Math Review

Module Summary
Students build the mathematical foundation necessary for automata theory. These concepts directly support the transition into finite automata in the next module.

Module 2: Deterministic Finite Automata (DFA)

Module Overview
This module introduces deterministic finite automata as the first formal model of computation used to recognize regular languages.

MLOs

- **MLO1:** Define and explain the structure of a DFA (**CLO1**)

- **MLO2:** Construct DFAs for simple languages (**CLO3**)
- **MLO3:** Analyze strings accepted by a DFA (**CLO3**)

Required Readings & Resources

- Hopcroft, Sections 2.1–2.2
- DFA examples and simulations

Topics

- DFA formal definition
- State transitions
- Language recognition

Assignments

- Quiz 2: DFA | Sept 6

Module Summary

Students learn how machines recognize patterns through deterministic models, forming the base for more flexible models in the next module.

Module 3: Non-Deterministic Finite Automata (NFA)+ Deterministic Finite Automata (DFA) Equivalence

Module Overview

This module expands computation models by introducing nondeterminism and proving equivalence between NFAs and DFAs.

MLOs

- **MLO1:** Explain nondeterminism in automata (**CLO1**)
- **MLO2:** Construct NFAs and convert them to DFAs (**CLO3**)
- **MLO3:** Demonstrate equivalence between DFA and NFA (**CLO3**)

Required Readings & Resources

- Hopcroft, Sections 2.3–2.6
- Conversion examples

Topics

- NFA definition
- Subset construction
- Equivalence proofs

Assignments

- Quiz 3: NFA | Sept 13
- **Checkpoint 1: Automata Review**

Module Summary

Students connect deterministic and nondeterministic models, solidifying understanding before moving to language representations.

Module 4: Regular Expression & Conversions

Module Overview

This module introduces regular expressions as an alternative way to represent regular languages and explores conversions between representations.

MLOs

- **MLO1:** Construct regular expressions for given languages (**CLO2**)
- **MLO2:** Convert between regex and automata (**CLO3**)
- **MLO3:** Explain pattern representation in computation (**CLO4**)

Required Readings & Resources

- Hopcroft, Chapter 3
- Regex tools/examples

Topics

- Regular expressions
- Regex ↔ automata conversions

Assignments

- Quiz 4: Regular Expressions | Sept 20
- **Midterm Exam (Modules 1–4)**

Module Summary

Students gain multiple perspectives for representing languages, bridging theory with practical tools like regex.

Module 5: Regular Languages & Equivalence

Module Overview

This module focuses on properties of regular languages, including closure properties and limits of finite automata.

MLOs

- **MLO1:** Apply closure properties to regular languages (**CLO3**)
- **MLO2:** Analyze equivalence across representations (**CLO2**)
- **MLO3:** Evaluate limitations of regular languages (**CLO3**)

Required Readings & Resources

- Hopcroft, Chapter 4

Topics

- Closure properties
- Pumping lemma (regular)
- Equivalence

Assignments

- Quiz 5: Regular Languages | Sept 27
- **Checkpoint 2: Cumulative Regular Languages**

Module Summary
Students understand what regular languages can and cannot represent, preparing for more powerful models.

Module 6: Context -Free Grammars (CFGs)

Module Overview
This module introduces grammars and context-free languages, emphasizing recursive structure and parsing.

MLOs

- **MLO1:** Define and construct CFGs (**CLO2**)
- **MLO2:** Derive strings using grammars (**CLO3**)
- **MLO3:** Interpret parse trees and ambiguity (**CLO3**)

Required Readings & Resources

- Hopcroft, Chapter 5

Topics

- CFG definitions
- Derivations
- Parse trees

Assignments

- Quiz 6: CFGs | Oct 4

Module Summary
Students move beyond regular languages into recursive structures critical for programming language design.

Module 7: Push Down Automata &Context -Free Language Properties

Module 7: PDA + CFL Properties

Module Overview
This module connects pushdown automata with context-free languages and explores their computational limits.

MLOs

- **MLO1:** Explain structure and operation of PDAs (**CLO1**)
- **MLO2:** Convert between CFGs and PDAs (**CLO3**)
- **MLO3:** Apply pumping lemma and closure properties for CFLs (**CLO3**)

Required Readings & Resources

- Hopcroft, Chapters 6–7

Topics

- PDA design
- CFG ↔ PDA

- CFL properties

Assignments

- Quiz 7: PDA & CFL | Oct 11

Module Summary

Students integrate grammar and automata, understanding how memory (stack) expands computational power.

Module 8: Turning Machines, Decidability & Church -Turing

Module Overview

This module introduces Turing machines as a general model of computation and explores the limits of computability.

MLOs

- **MLO1:** Explain Turing machine structure and computation (**CLO1**)
- **MLO2:** Analyze decidability and undecidability (**CLO3**)
- **MLO3:** Relate theoretical models to real-world computing systems (**CLO4**)

Required Readings & Resources

- Hopcroft, Chapters 8–9

Topics

- Turing machines
- Decidability
- Church-Turing thesis

Assignments

- Quiz 8: Turing Machines | Oct 16
- **Final Exam (Comprehensive)**

Module Summary

Students reach the theoretical limits of computation and connect automata theory to real-world systems and programming languages.